

RoadFury at the ICST 2026 Tool Competition – Self-Driving Car Testing Track

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Abstract—We present RoadFury, a test prioritization tool for the ICST 2026 SDC Testing Competition [?]. Existing approaches reduce road geometry to aggregate statistics, discarding the sequential structure that makes roads dangerous. RoadFury instead preserves the full geometric sequence: a 10-channel feature matrix encoding curvature, heading, jerk, and positional context at every road point, processed by a Pre-LN Transformer Encoder [?] stabilized with Stochastic Weight Averaging (SWA) [?]. Deployed as a Docker-containerized gRPC service, RoadFury achieves APFD = 0.804 ± 0.012 , surpassing the prior best of 0.781 and outperforming LLM, GNN, and CNN baselines.

Index Terms—test prioritization, self-driving cars, Transformer, stochastic weight averaging

I. INTRODUCTION

Testing self-driving cars (SDCs) in simulators such as BeamNG.tech is essential yet expensive [?], [?]. *Test prioritization* reorders a test suite so that failure-revealing cases execute first, maximizing early fault detection. The standard metric is APFD [?]:

$$\text{APFD}(\pi) = 1 - \frac{\sum_{i=1}^m TF_i}{n \cdot m} + \frac{1}{2n}, \quad (1)$$

where n is the total number of tests, m the number of failing tests, and TF_i the rank of the i -th fault in ordering π . A higher APFD indicates that faults are exposed earlier in the execution.

Existing tools such as SDC-Scissor [?], [?] summarize each road by three aggregate statistics (mean curvature, mean lateral position, standard deviation) and feed them into a feedforward network. This representation collapses the *ordered sequence* of road geometry into a fixed-size vector, discarding spatial transitions—consecutive sharp bends, sudden jerk changes—that most reliably predict SDC failures. **RoadFury** treats each road as a full sequence and learns over it with a Pre-LN Transformer Encoder [?] regularized by SWA [?].

II. APPROACH

A. Feature Extraction

Each test case is defined by a variable-length sequence of 2D road control points. We uniformly resample every road to $L=197$ points and compute 10 intrinsic geometry channels at each point (Table ??), yielding a (197×10) feature matrix that is translation- and rotation-invariant. All channels are z-normalized using per-channel mean and standard deviation computed on the training set.

TABLE I
10-CHANNEL ROAD GEOMETRY FEATURE SET

Ch.	Feature	Definition
f_0	Segment length	$\ p_{i+1} - p_i\ _2$
f_1	Abs. angle change	$ \Delta\theta_i $
f_2	Menger curvature	$\kappa_i = 4A_i/(a_i b_i c_i)$
f_3	Curvature jerk	$\Delta\kappa_i$
f_4	Cum. dist. norm	$\sum \ p_{j+1} - p_j\ /d_{\text{tot}}$
f_5	Heading sin	$\sin(\theta_i)$
f_6	Heading cos	$\cos(\theta_i)$
f_7	Rel. position	$i/(L-1) \in [0, 1]$
f_8	Local curv. std	$\text{std}(\kappa_{i-5:i+5})$
f_9	Curv. acceleration	$\Delta^2\kappa_i$

B. RoadTransformer Architecture

Fig. ?? illustrates the complete pipeline. A linear projection maps each 10-dim point vector to a 128-dim embedding, followed by LayerNorm and GELU activation. A learnable [CLS] token is prepended to the sequence and sinusoidal positional embeddings are added, yielding 198 tokens in total. A 4-layer Pre-LN Transformer Encoder [?] ($d=128$, 8 heads, FFN width 512, dropout 0.1) then applies global self-attention over the entire road trajectory. The resulting [CLS] representation passes through a two-layer MLP head (GELU activations, dropout) to produce a failure probability $\hat{y}$, which is used to rank tests in descending order. The model has 829K parameters in total.

C. Training and Deployment

We train on 28,804 labeled road–outcome pairs from SensoDat [?] using AdamW ($\text{lr}=5 \times 10^{-4}$, weight decay= 10^{-3}) with binary cross-entropy loss for 75 epochs. SWA [?] collects weight snapshots from epochs 50–75 and averages them, yielding a flatter loss landscape and improved generalization. The trained model is packaged as a Docker-containerized gRPC microservice exposing three endpoints: Name, Initialize, and Prioritize.

III. EVALUATION

A. Results

We evaluate on the 956-test competition set [?], averaging APFD over 30 independent random trials to account for tie-

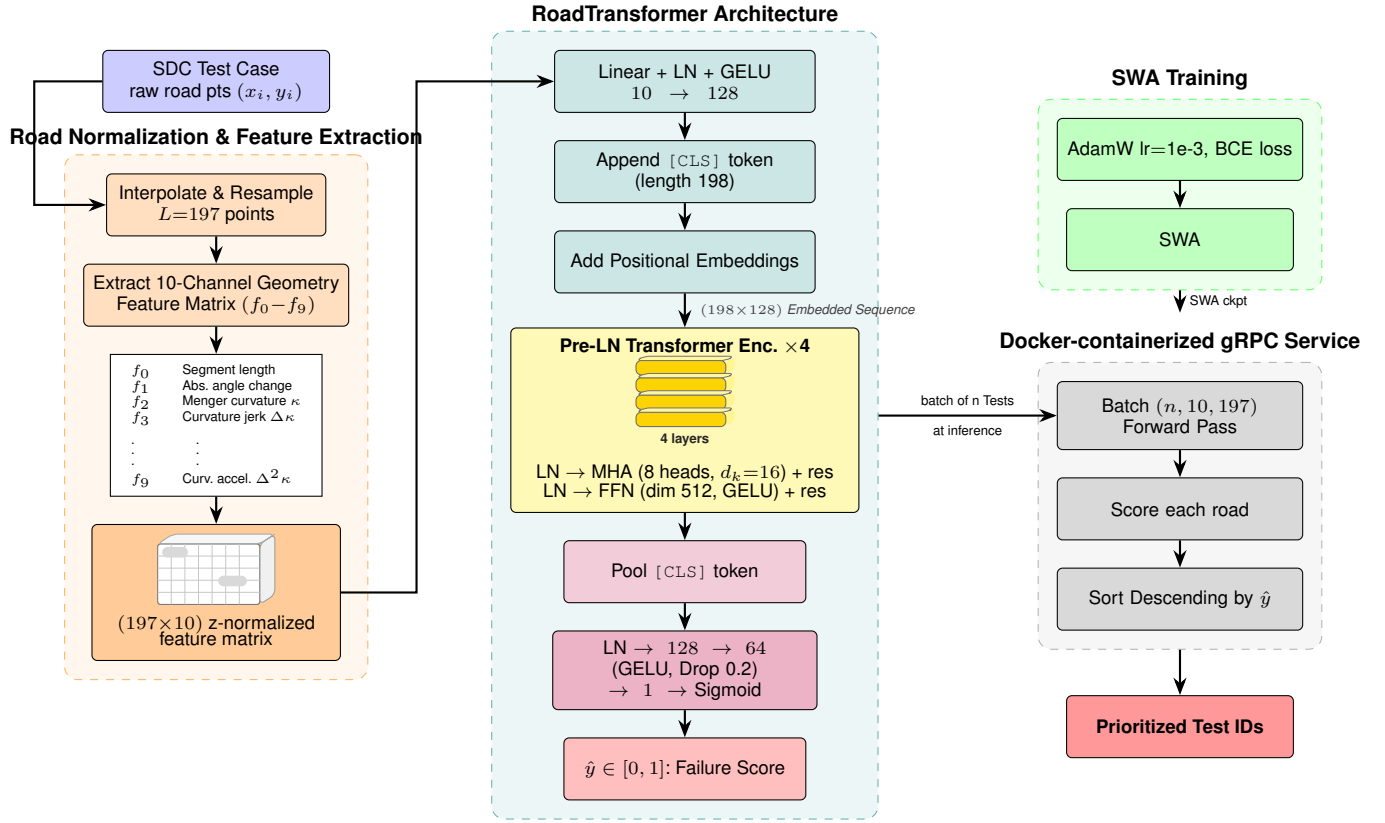


Fig. 1. RoadFury pipeline: (1) road control points are uniformly resampled; (2) 10 geometry channels are extracted per point; (3) a Pre-LN Transformer Encoder with a [CLS] token produces a failure score $\hat{y}$; the model is trained with SWA and served as a Docker gRPC service that returns prioritized test IDs.

TABLE II
MULTI-PARADIGM COMPARISON (956 COMPETITION TESTS, 30 TRIALS)

Method	APFD	Paradigm
Random	0.493	–
LLM zero-shot	0.487 ± 0.019	Language model
GNN (road graph)	0.533 ± 0.025	3-layer GCN [?]
ResNet-50 visual	0.572 ± 0.013	Road image CNN
ITEP4SDC	0.781	MLP, 3 features
RoadFury (ours)	0.804 ± 0.012	Transformer+SWA

TABLE III
ABLATION STUDY (956 COMPETITION TESTS, 30 TRIALS)

Modification	APFD
RoadTransformer (no SWA)	0.788 ± 0.014
+ Multi-scale Conv Stem	$0.767 \pm 0.014 \downarrow$
+ Stochastic Depth (DropPath)	$0.768 \pm 0.015 \downarrow$
+ Triple Pooling (CLS+Mean+Max)	$0.768 \pm 0.015 \downarrow$
+ Mixup augmentation	$0.773 \pm 0.015 \downarrow$
+ Focal Loss ($\gamma=2$) [?]	$0.782 \pm 0.014 \downarrow$
+ Test-time augmentation (TTA)	$0.770 \pm 0.016 \downarrow$
+ SWA	$0.804 \pm 0.012 \uparrow$

breaking variance in the ranking. Table ?? compares RoadFury against four alternative paradigms explored during development.

RoadFury outperforms all baselines. LLMs reason over natural-language road descriptions and miss fine-grained numeric geometry; GNNs capture pairwise adjacency but lose the ordered trajectory dynamics; CNNs applied to road images sacrifice the numeric precision of curvature values. RoadFury’s Transformer attends directly over ordered curvature transitions, capturing the spatial patterns most predictive of simulation failures.

B. Ablation

Table ?? reports the effect of seven modifications applied independently to the base RoadTransformer trained without SWA. All numbers are averaged over 30 trials on the 956-test competition set.

Every modification except SWA degraded or matched base performance. SWA is the only one that consistently improved APFD, confirming that weight averaging over flat optima matters more than architectural changes or data augmentation.

IV. CONCLUSION

Preserving sequential road geometry, rather than aggregating it, is the key to effective SDC test prioritization. A Pre-LN Transformer with SWA achieves $\text{APFD} = 0.804 \pm 0.012$, the best result in the ICST 2026 SDC Testing Competition. We plan to investigate richer simulation feedback signals and cross-simulator transfer. The tool is publicly available at <https://github.com/chisngyen/sdc-test-prioritization-novel>.

REFERENCES

- [1] C. Birchler, S. Khatiri, B. Bosshard, A. Gambi, and S. Panichella, “Machine learning-based test selection for simulation-based testing of self-driving cars software,” *Empir. Softw. Eng.*, vol. 28, no. 71, 2023.
- [2] C. Birchler, N. Ganz, S. Khatiri, A. Gambi, and S. Panichella, “Cost-effective simulation-based test selection in self-driving cars software with SDC-Scissor,” in *Proc. IEEE SANER*, 2022, pp. 386–390.
- [3] S. Yoo and M. Harman, “Regression testing minimization, selection and prioritization: A survey,” *Softw. Test. Verif. Reliab.*, vol. 22, no. 2, pp. 67–120, 2012.
- [4] P. Izmailov, D. Podoprikin, T. Garipov, D. Vetrov, and A. G. Wilson, “Averaging weights leads to wider optima and better generalization,” in *Proc. UAI*, 2018.
- [5] C. Birchler, S. Khatiri, C. Rohrbach, T. Kehrer, and S. Panichella, “SensoDat: Simulation-based sensor dataset of self-driving cars,” in *Proc. IEEE/ACM MSR*, 2024.
- [6] P. Aryan, T. Fulcini, L. Starace, C. Birchler, and S. Panichella, “ICST Tool Competition 2026 – SDC Testing Track,” in *Proc. IEEE ICST*, 2026.
- [7] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, “Focal loss for dense object detection,” in *Proc. IEEE ICCV*, 2017, pp. 2980–2988.
- [8] T. N. Kipf and M. Welling, “Semi-supervised classification with graph convolutional networks,” in *Proc. ICLR*, 2017.
- [9] R. Xiong *et al.*, “On layer normalization in the Transformer architecture,” in *Proc. ICML*, 2020.